ANALYSTLAB AFRICA

Experience Lab Internship Programme

HealthConnect Clinic
Machine Learning Problem Definition Document

By

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Dataset: HealthConnect Appointment Dataset (5,000 records)

Prepared for the AnalystLab Africa Experience Lab

Week 4 Submission – Data Science Track

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1. Introduction

This document presents the Data Science track's Week 4 output for the HealthConnect Experience Lab, prepared as part of the AnalystLab Africa Experience Lab Internship Programme. HealthConnect Clinic is a fictional outpatient healthcare provider experiencing a recurring operational problem: a substantial proportion of scheduled appointments are missed, creating inefficiency, wasted clinical capacity, and disruption to patient care.

The Week 4 brief frames the central project question as: “How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?” The Data Science track's specific responsibility within this shared project is to define the machine learning problem and assess whether the available appointment data can support a reliable no-show prediction solution.

This document reviews the resources provided for the Experience Lab, defines the machine learning problem, assesses the quality and suitability of the HealthConnect Appointment Dataset, proposes a target variable and candidate input features, and sets out an initial modelling approach together with the key considerations that should guide Week 5 development. Consistent with the Week 4 brief, this is a planning and problem-definition document: no model has been trained or deployed at this stage.

2. Review of Project Resources

Three project resources were reviewed for this track:

- **HealthConnect_Appointment_Data.csv:** 5,000 fictional, anonymised appointment records, comprising patient demographics, appointment details, booking information, prior appointment history, reminder information, distance to clinic, waiting time, and appointment outcome.
- **HealthConnect Data Dictionary:** Definitions of all 18 variables in the dataset, including data types, example values and notes on known limitations (for example, that reminder_channel and several other fields contain limited missing values).
- **HealthConnect Clinic Knowledge Base:** Approved fictional information about the clinic's locations, services, booking, rescheduling and late-arrival policies. This resource is primarily intended for the Generative AI track, but it was reviewed here because it usefully explains what a 'Cancelled' appointment represents operationally (a patient proactively contacting the clinic in advance), which directly informs how appointment outcomes are handled in Section 4.3.

The Appointment Dataset and Data Dictionary are the primary resources for this track. No project resources were altered; the analysis in this document was performed on a copy of the dataset, consistent with the brief's instruction not to overwrite original project resources.

3. Problem Definition

3.1. Business Problem

HealthConnect Clinic loses appointment capacity when patients fail to attend scheduled appointments without cancelling in advance. This reduces clinical efficiency, creates avoidable idle time, and prevents other patients from being offered those slots. The clinic wants to understand which patients are more likely to miss their appointments so that staff can intervene in advance, for example through additional reminders, phone confirmation calls, or proactive overbooking of slots identified as high-risk.

3.2. Machine Learning Problem Statement

The Data Science track frames this as a supervised binary classification problem: given information known about a patient and an appointment at the time of booking, predict whether that appointment is likely to result in a no-show. This framing is consistent with the wider literature on appointment no-show prediction, where binary classification using demographic, historical and scheduling variables is the dominant approach (Carreras-García et al., 2020; Dantas et al., 2018).

The model's intended use is to flag high-risk appointments in advance so that HealthConnect Clinic staff can prioritise outreach and scheduling interventions. It is not intended to replace clinic staff or automatically cancel or reallocate appointments.

4. Initial Data Assessment

4.1. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 rows and 18 columns, with one row per appointment. appointment_id is unique across all 5,000 rows and serves as the primary key. patient_id is not unique (1,696 distinct patients across 5,000 appointments), confirming that most patients in the dataset have more than one recorded appointment, which supports the inclusion of patient-level history variables such as previous_appointments and previous_no_shows.

The target field, appointment_outcome, takes three values: No-Show (2,423 records, 48.5%), Attended (2,314 records, 46.3%), and Cancelled (263 records, 5.3%). No duplicate rows were found.

4.2. Data Quality Assessment

- **Missing values:** reminder_channel is missing for exactly 1,366 records, which corresponds precisely to the 1,366 records where reminder_sent = 'No'. This is structural, expected missingness (no channel exists where no reminder was sent) rather than a data-quality problem, and should be encoded accordingly (for example, as a 'None' category) rather than imputed. distance_to_clinic_km (90 records) and waiting_time_minutes (60 records) have a small number of missing values that appear unrelated to appointment_outcome and will require a documented imputation strategy (for example, median imputation, consistent with the approach adopted in the Data Science track's Week 2 preprocessing work on a separate project) or exclusion, depending on the modelling approach chosen in Week 5.
- **Unusual pattern requiring clarification:** waiting_time_minutes is described in the Data Dictionary as an 'estimated waiting time' and is present for the large majority of No-Show (2,386 of 2,423) and Cancelled (261 of 263) records, with a distribution (mean approximately 24 minutes) that is statistically indistinguishable from the Attended group. This is difficult to reconcile with a literal in-clinic waiting time, since a patient who did not attend or who cancelled in advance would not generate an actual waiting period. Two explanations are plausible: either the field represents a pre-appointment estimate communicated at scheduling (in which case it may legitimately be usable as an input feature), or it is a value that would only be known during or after a clinic visit (in which case it cannot be used, as discussed in Section 6.2). This should be clarified with the data provider before Week 5 modelling begins.
- **Class balance:** The target classes are reasonably balanced between No-Show and Attended once Cancelled records are set aside (see Section 4.3), which is favourable for model training and reduces the need for resampling techniques that would otherwise add complexity at this early stage.
- **Data types:** booking_date and appointment_date are stored as text and will need to be parsed as dates. booking_lead_days was confirmed to equal the exact day difference between booking_date and appointment_date for all 5,000 records, so it is a derived, internally consistent field rather than an independent data-entry value.

4.3. Handling of Appointment Outcomes (Cancellations)

The brief specifically requires that this document define how appointment outcomes such as cancellations will be handled. Cancelled appointments (5.3% of the dataset) are operationally distinct from No-Shows: per the Clinic Knowledge Base, a cancellation reflects a patient proactively contacting the clinic in advance to release their slot, which is a materially different behaviour from silently failing to attend.

The proposed approach is to treat this as a binary classification problem restricted to Attended and No-Show records, with Cancelled records excluded from the primary training and evaluation dataset. This keeps the target variable behaviourally coherent (silent non-attendance versus attendance) and avoids conflating a proactive, clinic-friendly action with the problem the clinic is actually trying to reduce. As a documented alternative for consideration in Week 5, Cancelled and No-Show records could instead be combined into a broader 'non-attendance' class if the clinic's operational priority shifts towards slot-utilisation generally rather than no-show behaviour specifically; this trade-off should be revisited once stakeholder priorities are confirmed.

5. Proposed Target Variable

The proposed target variable is a derived binary field, referred to here as *is_no_show*, defined as 1 where appointment_outcome = 'No-Show' and 0 where appointment_outcome = 'Attended', calculated only on the subset of records that are not Cancelled (see Section 4.3). On this subset, the base rate is approximately 51% No-Show, which is a well-balanced target for classification.

6. Potential Input Features

6.1. Candidate Features and Supporting Evidence

An initial, exploratory review of no-show rates against several candidate variables was carried out to help prioritise features for Week 5. This is not a substitute for formal statistical testing, which is expected to form part of subsequent analysis, but it provides an early, evidence-based view of which variables appear most informative.

Candidate Feature	Type	Initial Evidence
previous_no_shows	Numerical (count)	Strong, near-monotonic relationship: no-show rate rises from 46% (0 prior no-shows) to 100% (5 prior no-shows).
booking_lead_days	Numerical	Strong relationship: no-show rate rises from 29% (≤7 days' notice) to 64% (30–60 days' notice).
reminder_sent	Categorical (binary)	Modest relationship: 49.9% no-show when a reminder was sent vs. 54.6% when it was not.
previous_appointments	Numerical (count)	Provides context for previous_no_shows; likely useful in combination (e.g., as a no-show rate ratio) rather than alone.
age / age_group	Numerical / categorical	Little variation observed across age groups (48%–53% no-show rate); may have limited standalone predictive value but retained pending multivariate analysis.
appointment_type	Categorical	Little variation observed (49%–54% across the four types); retained pending multivariate analysis.
distance_to_clinic_km	Numerical	Not yet tested against outcome; retained as a plausible predictor based on the wider literature (Dantas et al., 2018).
appointment_day / appointment_time	Categorical	Not yet tested against outcome; retained as candidate scheduling features.
gender	Categorical	Retained as a candidate demographic feature; to be evaluated jointly with a fairness review (see Section 8).

The early evidence that a patient's own history of missed appointments and the length of the booking lead time are the strongest available signals is consistent with the wider clinical literature on appointment no-shows, in which prior no-show history and longer lead times are repeatedly identified as the two most consistent predictors across specialties (Dantas et al., 2018; Liu et al., 2022).

6.2. Features Excluded from the Input Set

- **appointment_id and patient_id:** Unique or near-unique identifiers with no generalisable predictive meaning; excluded to prevent the model from memorising individual records.
- **appointment_outcome:** This is the source of the target variable and must not also appear as an input feature.
- **waiting_time_minutes:** Provisionally excluded from the candidate feature set pending clarification of what this field actually measures (see Section 4.2). If it is confirmed to be a value only known during or after the appointment, it must be excluded entirely, as including it would introduce data leakage: information that would not be available at the point a prediction is actually needed, before the appointment takes place (Kaufman et al., 2012).

All other fields are retained as candidate features pending the more detailed statistical and correlation analysis planned for Week 5.

7. Initial Modelling Approach

The following initial approach is proposed for Week 5 and beyond, and is intended to be refined once the data-quality questions in Section 4.2 are resolved:

1. **Data split:** Given that the dataset spans booking dates from November 2024 to June 2026, a chronological (time-based) train/test split is proposed rather than a purely random split, so that the model is evaluated on its ability to predict future appointments from past data, more closely mirroring real-world deployment (Liu et al., 2022).
2. **Baseline model:** Logistic regression is proposed as an interpretable baseline, consistent with its widespread use as a starting point in the no-show prediction literature (Carreras-García et al., 2020).
3. **Candidate models:** Tree-based ensemble methods, such as random forests or gradient boosting (e.g., XGBoost), are proposed as stronger candidates once a baseline is established, as these approaches have shown consistently strong performance in comparable appointment no-show studies (Chen & Guestrin, 2016; Dantas et al., 2018).
4. **Encoding and scaling:** Categorical variables (gender, appointment_type, appointment_day, appointment_time, reminder_channel) will require encoding, and numerical variables may require scaling depending on the final choice of algorithm.
5. **Evaluation metrics:** Given the clinic's interest in identifying at-risk appointments for intervention, precision, recall, F1-score and ROC-AUC are proposed as the primary evaluation metrics, rather than accuracy alone, since the cost of missing a true no-show (a false negative) is likely to matter more to the clinic than a false positive that simply triggers an extra reminder.
6. **Baseline comparison:** Model performance should be compared against a simple persistence baseline (for example, always predicting based on a patient's own historical no-show rate), which prior work has shown to be a meaningful benchmark in this problem domain (Liu et al., 2022).

8. Key Modelling Considerations

- **Data leakage:** As set out in Section 6.2, any variable that would not be known at the point of booking, or that could only be recorded during or after a clinic visit, must be excluded from the feature set (Kaufman et al., 2012).

- **Fairness and responsible use:** Because the model's predictions could influence how proactively different patients are contacted or supported, its performance should be reviewed across demographic subgroups (gender, age group) before use, consistent with the NIST AI Risk Management Framework's guidance on managing risk and promoting trustworthy AI (Tabassi, 2023). A model that is systematically less accurate, or that under- or over-flags a particular group, should not be deployed without mitigation.
- **Patient privacy:** Although the dataset is fictional and anonymised, any real-world equivalent would involve sensitive health-adjacent information; the modelling approach should be designed on the assumption that equivalent safeguards would be required in a production setting.
- **Reproducibility:** All data-cleaning and feature-engineering steps should be scripted and version-controlled, and any processed dataset should be saved separately from the original files, consistent with the Week 4 brief's resource-handling instructions.
- **Model as decision support, not automation:** Consistent with Section 3.2, the intended use of this model is to help clinic staff prioritise outreach, not to automatically cancel, reschedule or deprioritise a patient's appointment.

9. Assumptions, Limitations, Risks and Dependencies

- **Assumption – stability of behaviour:** The initial modelling approach assumes that the factors associated with no-shows in the historical data (2024–2026) will remain relevant for future appointments; this should be re-checked periodically once a model is in use.
- **Limitation – dataset is fictional and synthetic:** As a fictional, anonymised Experience Lab dataset, it may not fully capture the complexity or noise of a real clinical scheduling system, and any findings should be treated as illustrative of methodology rather than as clinical evidence.
- **Limitation – unresolved data-quality question:** The status of waiting_time_minutes (Section 4.2) is currently unresolved and represents a risk to the integrity of the feature set if not clarified before modelling begins.
- **Risk – class definition sensitivity:** The decision to exclude Cancelled records (Section 4.3) meaningfully shapes the problem definition; if HealthConnect Clinic's priorities differ from the assumption made here, the target variable and reported performance may need to be revisited.
- **Dependency – cross-track alignment:** This track's proposed target variable and feature set should be shared with the Machine Learning Engineering and Project Management tracks, since they depend on this problem definition to design the system architecture and project timeline respectively.

10. Week 4 Project Summary

- **Problem addressed:** Defining a machine learning problem to predict appointment no-shows at HealthConnect Clinic and assessing whether the available data can support it.
- **Resources used:** The HealthConnect Appointment Dataset (5,000 records) and Data Dictionary; the Clinic Knowledge Base was consulted to interpret the meaning of a 'Cancelled' outcome.
- **Key observations:** The dataset is largely clean and well-structured, with mostly structural or minor missing values; previous no-show history and booking lead time show the clearest early relationship with no-show behaviour; and one variable, waiting_time_minutes, shows a data-quality pattern that needs to be clarified before it can safely be used as a feature.

- **Proposed approach:** A binary classification problem (No-Show vs. Attended, with Cancelled excluded), starting from a logistic regression baseline and progressing to tree-based ensemble methods, evaluated using precision, recall, F1-score and ROC-AUC.
- **Key considerations:** Data leakage risk from waiting_time_minutes, the need for a fairness review across demographic subgroups, and the exclusion decision around Cancelled records.
- **Proposed focus for Week 5:** Resolve the waiting_time_minutes question with the data owner; complete feature encoding, scaling and a formal correlation/statistical review of the candidate features in Section 6.1; and train and evaluate the proposed baseline and candidate models.

References

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